SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

20/20

Annexure A

Industry Problem Solving Task

Code of Conduct:

(192421063)

I Neelakandamoorthy.C *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Neelakandamoorthy.c
192421063

Computer Networks

CSA0702

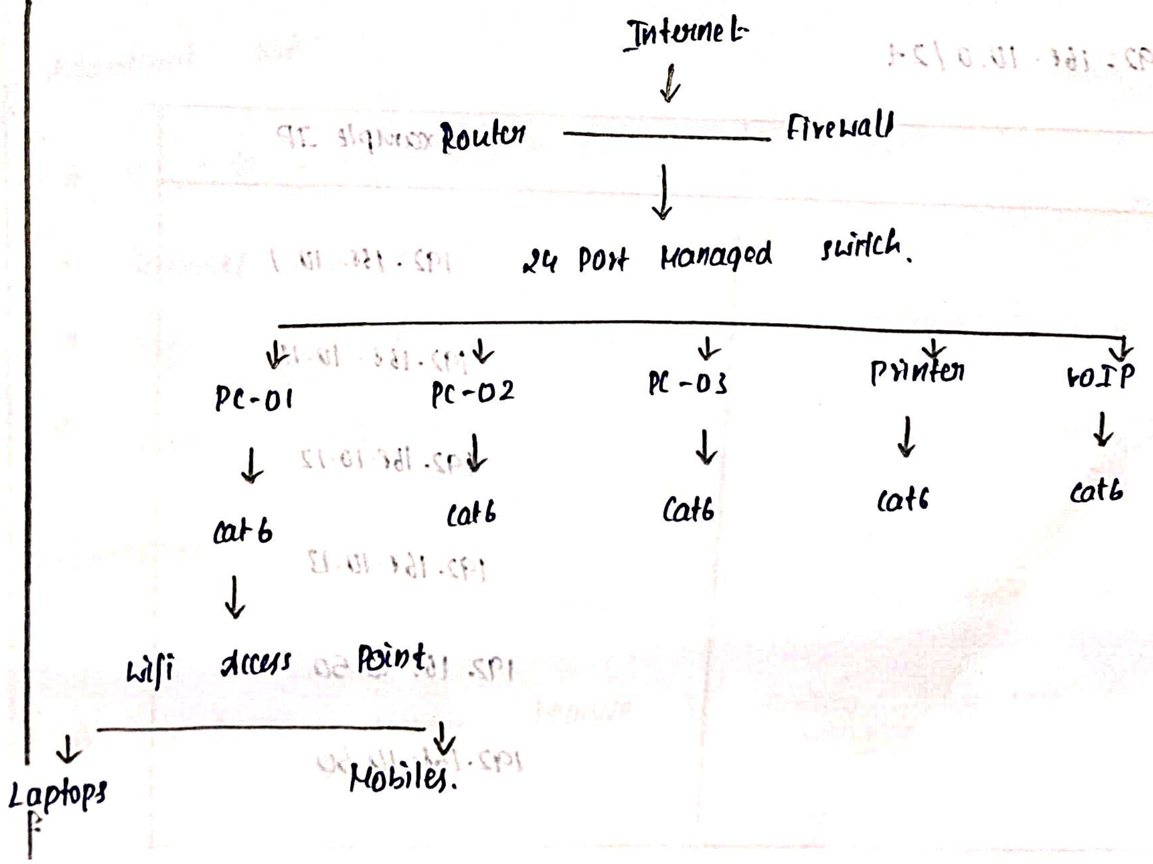
1) Problem Analysis :

A startup with 20 employees requires a reliable, easy to manage, and scalable network, each Employee needs access to the internet, shared files, Printers, VoIP services and wireless connectivity.

A star topology is selected because every end device connects independently to a central network switch.

Proposed Topology :

Star topology.



network Devices :

Device.	Quantity	Purpose
24 - Port Managed switch	1	central comm
Router / Firewall	1	security
wireless access point	1	wireless
Desktop PCs	20	Employee systems
Network printer	1	shared printers
VOIP Phones	As required	voice communication
cat6 UTP cables.	As required.	wired connections.

IP Addressing :

network : 192.168.10.0/24

Device	Example IP
Router	192.168.10.1
PC-01	192.168.10.11
PC-02	192.168.10.12
PC-03	192.168.10.13
Printer	192.168.10.50
access point	192.168.10.60

Advantages:

* Easy to install
* Easy trouble shooting
* Easy to expand
* Better network
* centralized Management

Testing and validation:

The network can be tested using:

Ping 192.168.10.1

Ping 192.168.10.12

Additional tests:

* PC - to - PC connectivity
* Internet connectivity
* Wifi connectivity
* Cable failure test.

Conclusion:

The star topology is the most suitable solution for a 20-person startup because it provides centralized Management. etc....

2)

Problem Analysis :

A banking organization requires extremely high availability beca
network failure can interrupt financial transaction and banking sem

A Partial Mesh topology is recommended at the core because Multiple redundant paths are avaible b/w critical network devices.

Proposed Topology :

Partial Mesh Topology.

Internet ·1

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Router 1

↙ ↘

Core Switch 1 - - - - - - - Core Switch 2

Firewall 1 - - - - - - - - - - Firewall 2

↓ ↓

Servers Servers.

↓

Distribution Switches

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Branch Networks.

Network devices :

Device	Purpose.
Enterprise Routers	Redundant internet / WAN connectivity
Layer 3 core switches	Routing b/w network segments
Firewalls	network security
Distribution switchs	Department / branch connectivity
servers	Banking application
UPS	Power backup.

IP Designs :

core network : 10. 10. 0. 0/16

server network : 10. 10. 10. 0/24

Management Network : 10. 10. 20/24

Banking Application network : 10. 10. 20. 0/24.

Conclusion :- A Partial Mesh Topology Provides the bank with high availability , redundancy , fault tolerance and rapid . recovery from network failures.

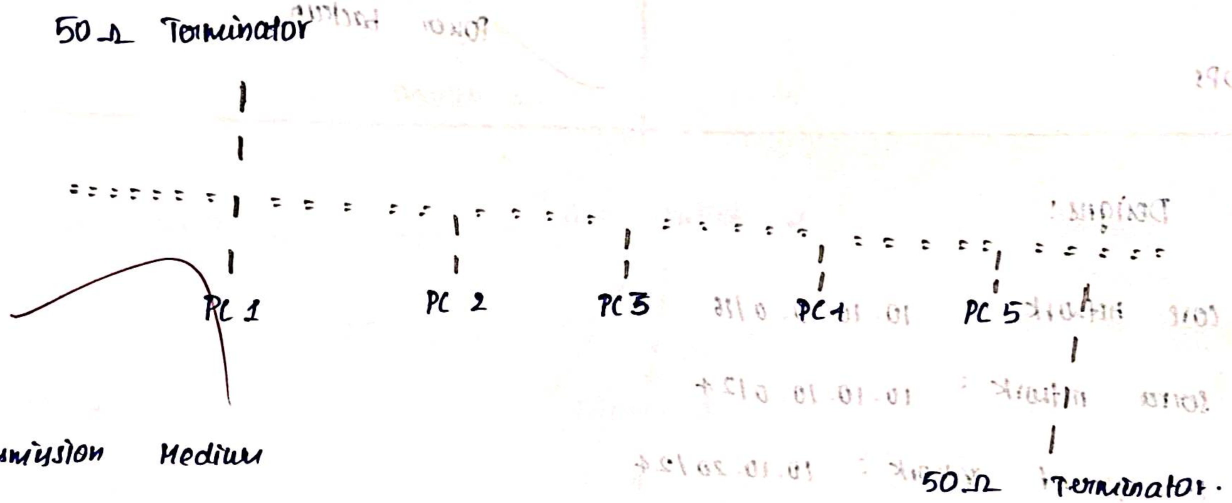
Problem Analysis :

A small classroom laboratory has a limited budget and relatively low network traffic. The network must provide basic connectivity for computers and shared resources.

A Bus Topology can be used as a low-cost conceptual solution

Proposed topology :-

Bus Topology



→ RG-58 coaxial cable.

* 50 -Ohm impedence
* uses BNC connectors.
* Ethernet bus networks

Network devices:

Device	Quantity	Purpose
Computers	As required	Student systems
Coaxial backbone	1	Shared communication
BNC -T- connectors	As required	Device connections
50 Ω Terminator	2	Signal transmission
Network Interface cards	As required.	Network.

Testing :

Perform : ping 192.168.30.12

* PC - to - PC communication
* File sharing
* Internet access
* Backbone failure.

Conclusion : A Bus topology can satisfy the requirements of a small, low budget class room environment, although modern installation would normally prefer a switched star topology.

1)

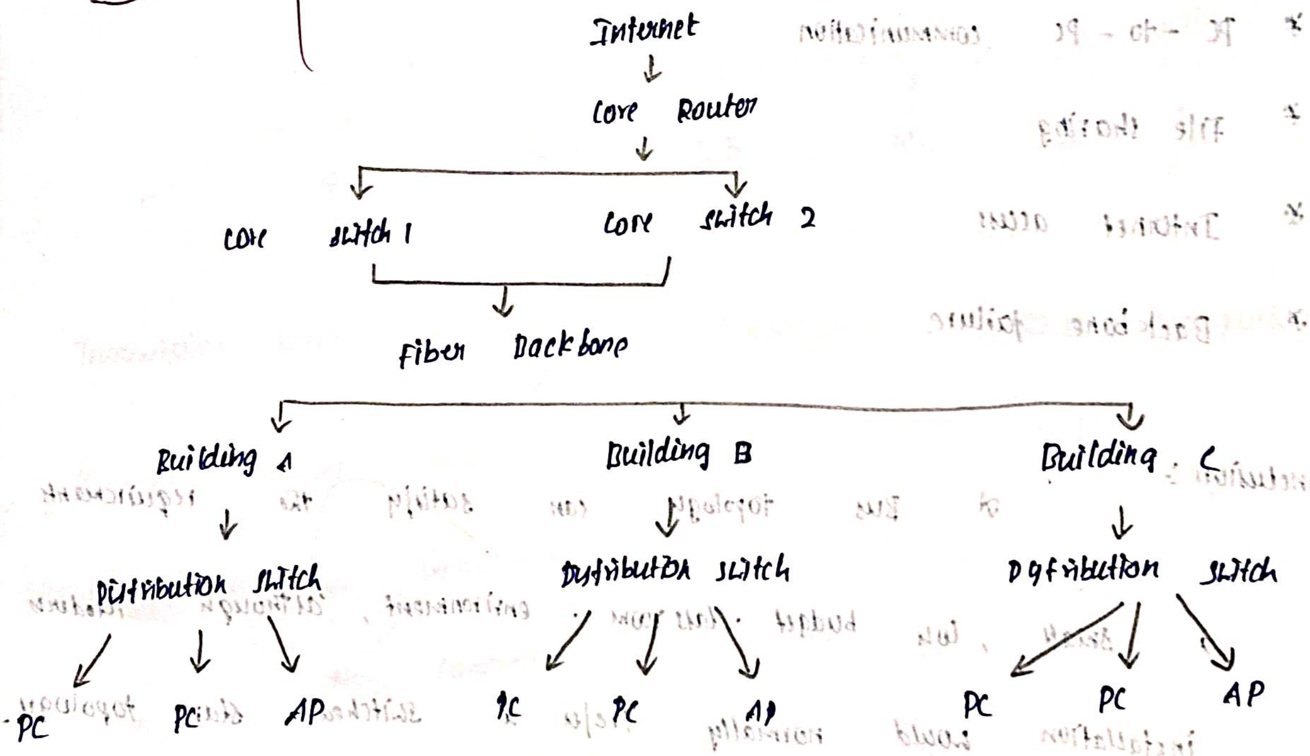
Problem analysis:

A university campus contains thousands of users distributed across multiple buildings such as:

* Engineering blocks
* Libraries
* Laboratories
* Hostels
* Administration buildings
* Lecture halls.

Therefore, a Hybrid topology is recommended.

Proposed Topology: Hybrid Topology.



Wireless network:

Access points can be connected using PoE.

This allows both Data and Electrical Power to travel through the Ethernet cable.

Conclusion:

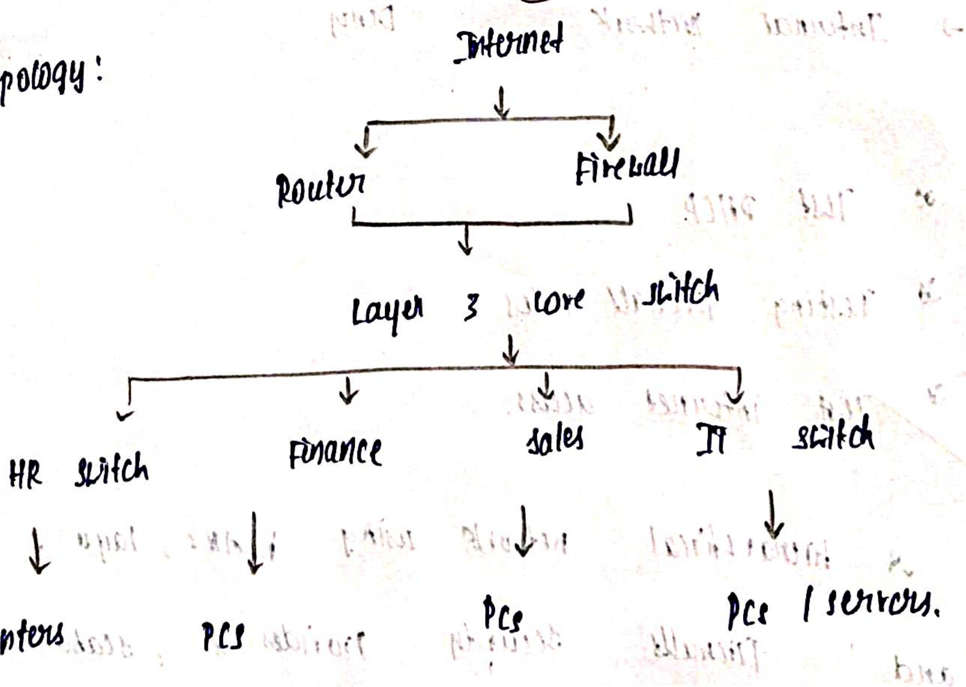
A Hybrid Topology is the best solution for a large university because it combines redundancy, scalability, centralized Management, cost effective access networks.

5) Problem analysis:

A company containing HR, Finance, sales, IT, and other departments requires secure communication while preventing unnecessary broadcast traffic.

A Hiearchical star /Tree architecture using VLANs is recommended.

Proposed Topology:



Inter - VLan Routing :

The Layer 3 switch Performs Routing b/w VLANS.

HR VLAN

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192.168.10.0/24

↓

Layer 3 switch

↓

Finance VLAN

↓

192.168.20.0/24

For Eo :

HR → HR server Allow

Finance → finance server Allow

Sales → HR database Deny

Guest → Internal Network Deny

Testing :

* Test DHCP
* Testing firewall rules
* Test Internet access.

Conclusion :

A hierarchical network using VLANs, layer 2 switches, Layer 3 routing, and firewalls security provides a scalable and secure solution for a Multi - department organization.